Fiscal Year:	FY 2018	**Task Last Updated:**	FY 07/10/2018
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Project Title:	Differential Effects on Telomeres and Telomerase in Twin Astronauts Associated with Spaceflight		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
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Task Description:	The ends of human chromosomes are capped by telomeres, tandem arrays of repetitive DNA sequence, which together with associated proteins serve to protect chromosomal termini from inappropriate degradation. Functional telomeres also prevent these natural DNA ends from being recognized as broken DNA (double-strand breaks--DSBs) and triggering inappropriate DNA damage responses, which could result in chromosomal instability. It is well established that telomere length diminishes with cellular division and normal aging. It is also becoming increasingly appreciated that telomere length is influenced by a variety of other factors including life stress, infection, inflammation, and oxidative stress. Nobel Laureate Elizabeth Blackburn suggests that telomere shortening “powerfully quantifies life’s insults.” Indeed, recent research supports telomere length maintenance as a key integrating component for the cumulative effects of genetic, environmental, and lifestyle factors on aging and aging-related diseases. Thus, the rate at which telomeres shorten provides an informative biomarker of biological aging and further, erosion of telomere length can be linked to age-related pathologies including reduced immune function, loss of fertility, cardiovascular disease, and cancer. Healthy lifestyles and diets are positively correlated with telomere length, and changes in diet and lifestyle have also been shown to modulate telomerase activity. Telomerase is the specialized enzyme present in germ line, stem, and cancer cells, capable of de novo addition of telomeric repeats onto the ends of newly synthesized telomeres. A large remaining puzzle in human populations is the relationship between telomerase activity (e.g., measured in peripheral blood lymphocytes) and lifestyle factors (e.g., nutritional, psychological, and physical stressors). We speculate that for the astronauts, telomere length represents a particularly relevant biomarker, because it reflects the combined exposures and experiences encountered during space travel. That is, an individual astronaut’s genetic susceptibilities, exposures to galactic cosmic radiation, physical, and psychological stressors encountered, are all “captured” as changes in telomere length. A remarkable once-in-a-life time opportunity exists to evaluate the effects of spaceflight factors on the monozygotic twin astronauts Scott and Mark Kelly, who will have spent a total of 540 and 54 days in low Earth orbit, respectively. We propose to assess differential effects on telomere length and telomerase activity associated with the upcoming yearlong International Space Station (ISS) mission in the twins – individuals matched genetically and who share similar, unique lifestyles; i.e., they are of similar “nature and nurture.” Standard assays will be employed to determine telomere length and telomerase activity as part of their respective and concurrent biodosimetry/chromosome analyses pre-flight (to establish baselines), during flight (to evaluate short-term/temporary changes), and post-flight (to evaluate long-term/permanent changes). Results from the space- and ground-based twins will be compared to evaluate effects of spaceflight factors on these informative biomarkers. Results from the twins will also be compared with our current retrospective analyses of a cohort of unrelated astronauts who have participated in missions of 3 to 6 months duration to evaluate inter-individual sensitivity to these effects of spaceflight, a strategy that will also facilitate evaluation of time dependent responses.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Aging and age-related degenerative pathologies like dementia, cardiovascular disease, and cancer are everyday concerns on Earth as well as in space. Taken together with the other TWINS Study investigations, results from this research will facilitate identification of interactive effects and relationships that will suggest potential mitigation strategies for future study, as well as benefit personalized medicine efforts in monitoring general health, and assessing aging and disease risk not only of astronauts, but for those on Earth as well.
Task Progress:	Telomeres are repetitive features of chromosomal termini essential for maintaining genomic integrity; they protect physical DNA ends from degradation and prevent them from triggering inappropriate DNA damage responses (DDRs). Telomere length shortens with cell division and thus with age, as well as with a host of lifestyle factors (e.g., stress) and environmental exposures (e.g., air pollution, radiation). Here, average telomere length was evaluated before, during, and after flight [peripheral blood mononuclear cells (PBMCs)] utilizing quantitative real-time polymerase chain reaction (qRT-PCR). Consistent with a strong genetic component, both twins had relatively similar telomere lengths before flight, and telomere lengths for the ground subject (HR) remained relatively stable for the duration of the study. Most striking, and contrary to expectation, was a significant increase in telomere length during flight for the flight subject (TW), as compared to his pre- and post-flight data, as well as to HR. Interestingly, telomere length shortened rapidly upon TW’s return to Earth – within ~48 hours – then stabilized to near preflight levels within months. As an independent measure, metaphase chromosomes were also prepared from pre-, in-, and post-flight samples, and thousands of individual telomeres evaluated utilizing telomere fluorescence in situ hybridization (Telo-FISH). Importantly, Telo-FISH cell-by-cell analyses facilitated generation of telomere length distributions, which confirmed increased telomere length during spaceflight, validating the qRT-PCR results. Upon return to Earth, TW’s telomere length distributions shifted back towards increased numbers of shorter telomeres. Telomerase is the reverse transcriptase capable of de novo addition of telomeric repeats onto the ends of newly replicated chromosomes. Telomerase activity was evaluated before, during, and after flight utilizing the qRT-PCR Telomeric Repeat Amplification Protocol (TRAP). In general, TW had lower levels of telomerase activity than HR throughout the study, whereas levels for each individual were similar pre- and post-flight. Unfortunately, due to unavoidable transit time and/or temperature associated with return, telomerase activity was lost in all inflight ambient samples collected onboard the ISS. Chromosome aberrations were analyzed to evaluate potential telomere-related instability (e.g., fusions), as well as to assess DDRs to ionizing radiation (IR) exposure during spaceflight; specifically, galactic cosmic ray (GCR)-induced cytogenetic damage. High-resolution directional genomic hybridization (dGH) paints for chromosomes 1, 2, and 3 were employed to enable, for the first time in astronauts, simultaneous detection of inter-chromosomal translocations and intra-chromosomal inversions. TW’s increased inversion frequencies were consistent with reported inflight IR exposure, particularly to high linear energy transfer (LET) space radiations. Mechanistic insight into these intriguing results was pursued via data sharing between other Twins Study investigations, and results are soon to be published.
Bibliography Type:	Description: (Last Updated: 12/10/2025)

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